ArtBench-10 Generative Modeling: A Comparative Study of VAE, GAN, and Diffusion Models

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Abstract. Nothing yet.

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VAE

DCGAN

Diffusion Models

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Variational Autoencoders (VAE)

Generative Adversarial Networks (GAN)

Diffusion Models

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4.1 Hyperparameter Optimization Strategy

We performed hyperparameter optimization with Bayesian optimization (Optuna). To keep the search feasible on standard hardware, tuning was conducted on a development subset comprising 20% of the training data. For each model

family, candidate configurations were trained on a training split and selected using performance on a held-out validation split. We minimized the following surrogate objectives: (i) *VAE*: validation reconstruction loss; (ii) *GAN*-based models: validation sum of discriminator and generator losses; (iii) *Diffusion models*: validation denoising loss under a fixed Gaussian noise schedule. The hyperparameter search spaces are summarized in Table 1, and the best configurations found on the studies are reported in Table 2. We used validation losses as surrogates during tuning because FID/KID are computationally expensive and high-variance at the per-trial scale: they require generating large sample sets and repeated feature extraction. In contrast, validation losses are available at each epoch and exhibit lower variance which improves sample efficiency under limited computational resources. We note, however, that GAN losses are far from beeing perfect proxies for perceptual sample quality; therefore, we use this objective only for hyperparameter selection during tuning and rely on qualitative inspection for the selected final parameters.

Table 1. Hyperparameter search spaces used in Bayesian optimization.

Model	**Parameters**	**Values / Range**
VAE	latent_dim, base_channels, batch_size	$\{16, 32, 64\}$; $\{16, 32, 64\}$; $\{32, 64, 128\}$
	lr, β	$[10^{-4}, 3 \times 10^{-3}]$ log-uniform; $[10^{-6}, 10^{-3}]$ log-uniform
DCGAN	latent_dim, feature_maps, batch_size	$\{128, 256, 512\}$; $\{32, 64\}$; $\{32, 64, 128\}$
	lr, β_1, real_label	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $[0.3, 0.7]$; $[0.85, 1.0]$
PixelUNet	model_channels, batch_size, num_timesteps	$\{32, 64, 96\}$; $\{16, 32, 64\}$; $\{500, 1000\}$
	lr	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform
LatentDenoiser	model_channels, num_res_blocks, batch_size	$\{32, 64, 96\}$; $\{2, 3, 4\}$; $\{16, 32, 64\}$
	lr, num_timesteps	$[10^{-4}, 5 \times 10^{-4}]$ log-uniform; $\{500, 1000\}$

4.2 Evaluation Protocol

Across the selected configurations (Table 2), the optimization process consistently favored smaller batch sizes. This aligns with the well-documented generalization gap observed in large-batch training [2]. Specifically, the inherent noise in small-batch stochastic gradients serves as a form of implicit regularization,

facilitating the escape from sharp local minima and leading the model toward flatter, more robust regions of the loss landscape. Regarding the learning rates, the observed values are relatively low, consistent with the need for stable convergence and stability during training, especially for GANs which are known to be sensitive to hyperparameter settings. Latent dimensions reveal important differences across model families: the VAE favored a much smaller latent space (32) compared to the DCGAN (256), which may reflect the different in the model's architectures: GANs don't pay an explicity penalty (KL divergence) for using larger latent spaces, while VAEs are encouraged to find compact representations. A larger latent space may encourage more diverse samples as long as the generator actually uses it. For the VAE, the optimal β value is very low, indicating that the model prioritizes reconstruction fidelity over latent space regularization. For the DCGAN, the optimal β_1 value is around 0.48, which is close to the commonly used value of 0.5 [5], suggesting a balance between stability and convergence speed. The real_label parameter being close to 1 (0.9836) indicates that the discriminator benefits from a small anti-overconfidence margin, which can help prevent it from becoming too confident and overpowering the generator during training without weakening the learning signal too much.

Table 2. Best configurations obtained.

Parameter	**VAE**	**DCGAN**	**PixelUnet**	**LatentDenoiser**
Best objective value	0.02658	1.68158	0.0358	0.2598
Learning rate	0.0005317	0.0001261	0.0004748	0.0001769
Batch size	32	32	16	32
Latent dimension	32	256	64	64
Base channels / feature maps	64	64	64	64
β (VAE)	1.14×10^{-6}	–	–	–
β_1 (Adam, DCGAN)	–	0.4813	–	–
real_label (DCGAN)	–	0.9836	–	–
num_timesteps (diffusion)	–	–	1000	1000
num_res_blocks (LatentDenoiser)	–	–	–	4

5 Results and Interpretation

5.1 Quantitative Results

Table 3 summarizes the quantitative evaluation across model families using 5,000 generated samples and 10 random seeds per model. We report mean ± standard deviation across seeds. Arrows indicate whether lower (↓) or higher (↑) values are preferred. The best value in each metric is highlighted in bold.

Table 3. Summary of quantitative results (mean ± std across seeds).

Model	FID ↓	KID ↓	Precision ↑
VAE	134.228 ± 0.531	0.13136 ± 0.00109	0.80548 ± 0.00671
DCGAN	**101.248 ± 0.864**	**0.07664 ± 0.00153**	0.86400 ± 0.00489
PixelUNet	116.933 ± 0.708	0.09972 ± 0.00107	**0.90804 ± 0.00512**
LatentDenoiser	119.271 ± 0.416	0.10068 ± 0.00122	0.78472 ± 0.00466

Overall, DCGAN achieves the best performence in terms of global distributional alignment obtaining the lowest FID (101.248 ± 0.864) and KID (0.07664 ± 0.00153) scores. This suggests that the DCGAN is more effective at capturing the overall data distribution compared to the other models. The VAE, on the other hand, has the worst performance in terms of FID and KID, which is consistent with the known limitations of VAEs in generating overly smooth reconstructions due to the variational objective, which can lead to blurry samples and less realistic outputs. PixelUNet achieves the highest precision (0.90804 ± 0.00512), indicating that it generates a higher proportion of samples that are close to the real data manifold. However, the higher FID and KID values compared to DCGAN suggest that the improvement in sample realism does not necessarily translate to better overall distributional alignment. They, therefore, generate more realistic samples but with less diversity. The LatentDenoiser achieves intermediate FID and KID values but the lowest precision among the evaluated models. This suggests that while its generated distribution is moderately aligned with the target dataset at a global level, a larger fraction of its individual samples fall outside the real data manifold compared to the other architectures. Such behavior may indicate incomplete convergence or reduced stability in the denoising process during sampling. In addition, the small standard deviations across seeds for all models suggest that the results are consistent and not heavily influenced by random initialization or stochastic training dynamics which is a positive sign of the robustness of the training process.

6 Conclusion

Use of External Resources and AI Tools

This project utilized ArtBench-10 (Kaggle). Gemini assisted in LaTeX formatting and analysis verification.

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